Sunmathi S

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EDUCATION

- **Presidency College of Arts and Science** — Chennai
  Master of Computer Applications; CGPA: 8.9 — *August 2024 - April 2026*
  Courses: *Web Technology, Data Structures, Analysis Of Algorithms, Artificial Intelligence, Machine Learning, Networking, Databases*
- **PSG College of Arts and Science** — Coimbatore
  Bachelor of Science in Mathematics; CGPA: 8.9 — *June 2021 - May 2024*
  Courses: *Linear Algebra, Graph Theory, Real Analysis, Complex Analysis, Dynamic and Static Mechanics , Operations Research*

SKILLS SUMMARY

Languages	Python, JavaScript, SQL, HTML5, CSS3
Frameworks	FastAPI, React.js, Three.js, Tailwind CSS
Backend	RESTful APIs, JWT Authentication, RBAC, WebSockets, Alembic
Database	PostgreSQL — schema design, query optimisation
Tools	Git, GitHub, VS Code, npm
Soft Skills	Leadership, Public Speaking, Event Management, Technical Writing

INTERNSHIPS

- **SpaceTug** — Onsite
  Web Developer — *December 2025 - April 2026*
  - **Backend Development**: Designed and implemented RESTful APIs using Python and FastAPI, enabling core application features including data retrieval, processing pipelines, and service integration across the platform.
  - **Frontend Development**: Built and maintained React.js components integrated with backend services via REST APIs.
  - **Authentication System**: Implemented JWT-based authentication and role-based access control to enforce secure user sessions, token validation, and protected route handling across the application.
  - **Database Management**: Designed schemas and wrote optimized PostgreSQL queries for efficient data storage and retrieval.
  - **API Integration**: Connected frontend interfaces with backend endpoints, ensuring smooth data flow, proper error handling, and consistent response formatting across all application modules.
- **Sri Senthil Aandavar Polysacks** — Onsite
  Web Developer — *May 2026 - June 2026*
  - **Frontend Development**: Built responsive web pages using HTML, CSS, and JavaScript, ensuring compatibility across devices and screen sizes with clean, maintainable code structure.
  - **Business Website**: Developed a complete business profile website featuring structured content layout, service showcases, and interactive UI elements tailored to represent the company's brand and offerings effectively.
  - **Database Integration**: Worked with databases to store, manage, and retrieve website content dynamically.
  - **Performance & Usability**: Assisted in cross-browser testing, UI refinements, and performance optimization to improve overall site responsiveness and user experience.

PROJECTS

- **Satellite Collision Avoidance System (Python, FastAPI, PostgreSQL, React.js, WebSockets)**: Real-time system that tracks satellites in orbit, predicts orbital conjunctions, and alerts users of potential collision risks. Ingested and propagated TLE orbital data using modular backend services built with FastAPI, performed conjunction analysis with configurable risk threshold evaluation, and triggered automated alerts for high-risk close approach events. Designed a PostgreSQL schema for satellite catalog management, integrated Alembic for database migrations, and implemented a scheduling engine for periodic propagation and screening tasks. Streamed live satellite position updates to a React.js frontend featuring 3D globe visualization via WebSocket connections, and generated structured PDF reports for mission analysis and documentation.
- **RPOD Mission Simulation System (Python, FastAPI, React.js, Three.js, NumPy/SciPy, WebSockets)**: Real-time spacecraft mission simulation system modeling orbit transfer, rendezvous, proximity operations, and docking with a target space station. Implemented orbital mechanics including Hohmann transfer and phasing orbit calculations, relative motion modeling using Clohessy-Wiltshire equations, and a full GNC pipeline comprising EKF-based navigation, guidance logic, and MPC/PID trajectory control. Built a constraint engine managing fuel limits and burn thresholds, a safety system enforcing. Streamed real-time telemetry to a React.js frontend with Three.js 3D spacecraft visualization via WebSockets.
- **Sri Senthil Aandavar Polysacks - Business Profile Website (HTML, Tailwind CSS, JavaScript)**: Designed and developed a fully responsive business showcase website for Sri Senthil Aandavar Polysacks, featuring a structured multi-section layout covering company overview, product listings, and contact information. Implemented interactive UI elements and smooth navigation using vanilla JavaScript, and applied Tailwind CSS utility classes for consistent, mobile-friendly styling across all devices.

CERTIFICATIONS

- **NPTEL Swayam Certification**: Soft Skills Development (IIT/NIT consortium) — Completed a nationally recognised online certification covering professional communication, interpersonal skills, and workplace effectiveness.
- **Tanii Hackathon**: Participated in the TANII Hackathon and completed a project-based learning experience.